The Use of Digital Twins in the Energy Sector: Selection and Verification of Tools for Modeling Hybrid Energy Supply Systems Based on Renewable Energy Sources

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***Abstract*—The aim of the work is to develop and test a set of measures to improve the efficiency of projects for the construction of hybrid energy complexes with power plants based on renewable energy sources by using digital twins. The paper shows the effectiveness of research and development of recommendations on the choice of parameters and modes of operation of hybrid power complexes having a diverse composition of generating plants and high dependence on natural uncertain factors at all stages of the life cycle using the developed universal digital twin, which was verified by the example of real power complexes with different equipment composition.**

Keywords—renewable energy, hybrid energy complex, energy storage system, wind power plant, photovoltaic plant, digital twin

I. INTRODUCTION

The use of power plants based on renewable energy sources for uninterruptible power supply to consumers in remote and isolated energy regions is ensured by the use of various methods for obtaining guaranteed capacity, including the combination of one or more types of renewable energy power plants into hybrid energy complexes (HECs) together with duplicate sources of guaranteed power supply on organic fuel and (or) energy storage systems [1 – 9].

In comparison with diesel generation, the introduction of such HECs allows for a significant reduction in the consumption of long-distance imported organic fuel, a reduction in the cost of energy production and an improvement in the local environmental situation, while ensuring a guaranteed energy supply to consumers.

Currently, there is no comprehensive methodology for creating HECs with renewable energy installations and energy storage systems.

The creation of physical models of each of the considered variants of HECs is difficult, expensive and time-consuming. An alternative to the creation of physical models is the use of "digital twins" - simulation digital models containing historical and the most current data on a physical object or process, which allows optimizing the parameters and operating mode of a HEC with constant verification of the model based on feedback from physical objects.

A considerable amount of work has been carried out both in our country and abroad devoted towards the issues concerning modeling and optimization of HECs using renewable energy. Based on these works, various digital twins have been created, allowing to solve problems related to modeling and optimization of parameters of HECs with generation based on renewable energy sources.

Therefore, an important task is the analysis of verification of modern digital twins for modeling HECs using actual data from the operation of real energy complexes in order to determine the most effective and universal modeling tool [10 – 14].

The selected verified digital twin can be used to correctly assess the effectiveness of various support measures at the project's initiation stage.

II. THE USE OF DIGITAL TWINS IN THE ENERGY SECTOR

At present, the concept of "digital twin" has a large number of interpretations [10 – 14]. With regard to the energy sector, a digital twin is a real display of all components of an object / system in the life cycle using physical data, virtual data and data of interaction between them. In other words, a digital twin is a virtual prototype of a real object/system, which can be used to conduct experiments and test hypotheses with high accuracy, predict the behavior of an object/system, and solve problems of managing its life cycle [14].

The main advantages of using digital twins to calculate the parameters and operating modes of the HECs based on renewable energy sources include:

1. **Economic efficiency:** Digital twins enable detailed economic analysis, including capital and operating cost calculations, as well as assessment of the cost of electricity, which helps to identify the most cost-effective solutions and optimize investments in renewable energy technologies.

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2. **Simplifying the design process:** Digital twins significantly simplify the design process of the HECs with different types of renewable energy plants and energy storage systems, allowing for the rapid generation and evaluation of multiple energy system configurations and the selection of the best technical solutions for specific conditions and requirements.

3. **Sensitivity analysis:** Digital twins enable sensitivity analysis, which helps assess how changes in key parameters (such as fuel prices, solar radiation levels, or wind speeds) affect the performance, resilience, and economics of the energy system.

4. **Simulation of different scenarios:** Digital twins allow simulating different scenarios of operation of hybrid energy complexes based on renewable energy sources, including seasonal changes, fluctuations in energy consumption and resource availability, which makes it possible to assess the resilience of the system and its ability to cope with different conditions.

5. **Support for operational decision making:** Digital twins help users better understand the characteristics and performance of renewable energy sources, which improves and accelerates the decision-making process, enabling the creation of informed strategies for the implementation and operation of renewable energy installations.

6. **Conducting virtual experiments:** Digital twins allow experiments and tests to be conducted in a virtual environment under operating conditions close to real ones, which significantly saves time and resources and makes it possible to explore various operating modes of equipment without the need for significant capital investments and risks associated with real equipment operation.

The article [15] provides a detailed analysis of the functional capabilities, advantages and disadvantages. Based on the analysis, it was found that HECs based on renewable energy sources can be most fully modeled and studied using digital twins such as HOMER Pro, HOGA and RES Energy Complex.

The HOMER Pro digital twin was developed at the National Renewable Energy Laboratory and improved by HOMER Energy (USA) (Figure 1) [16].

It is designed to solve problems of structural optimization of HEC with the determination of the optimal ratio of the parameters of the main generating equipment. Optimization is performed using a special patented algorithm called the HOMER Optimizer. This algorithm is proprietary, so users do not have access to it. The target functions can be capital investments, the volume of consumption of organic fuel and greenhouse gas emissions [16].

HOMER Pro is currently the world's most widely used digital twin for optimizing the design of HECs that include the use of renewable energy generation.

The digital twin HOGA (Spain) is designed for modeling and optimizing HEC with power plants based on renewable energy sources, including those connected to an external electrical network (Figure 2).

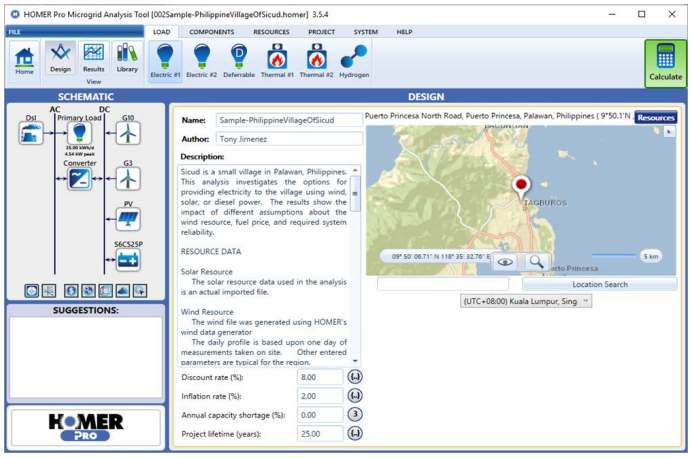


Fig. 1. HOMER Pro Digital Twin Interface.

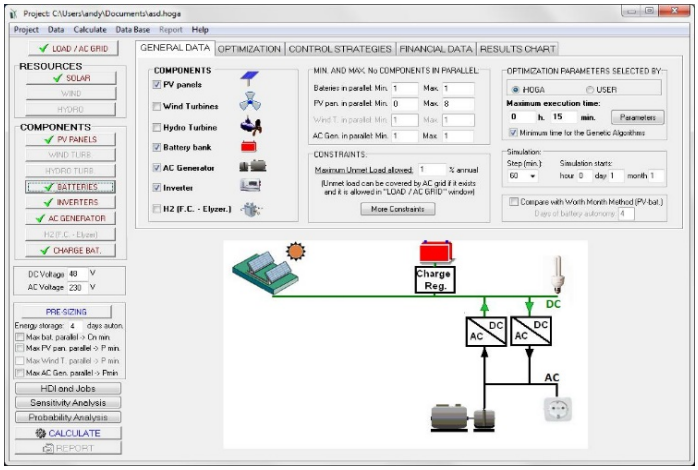


Fig. 2. HOGA Digital Twin Interface.

Optimization of the parameters of HEC is achieved by minimizing the total costs (or maximizing profits) throughout its life cycle, i.e., in general, the optimization is economic. The digital twin also allows for multi-objective optimization, which takes into account not only economic criteria, but also the simultaneous minimization of other variables selected by the user (for example, equivalent greenhouse gas emissions). When solving the optimization problem, a genetic algorithm is used to search for the most adapted individual, capable of withstanding external and internal disturbances [17].

The digital twin of RES Energy Complex was developed at the Department of Hydropower and Renewable Energy Sources of the National Research University "MPEI" (Russia) (Figure 3).

RES Energy Complex allows to quickly and accurately determine the optimal composition, parameters and modes of operation of HECs with power plants based on renewable energy sources for any time interval and territory specified by the user. When solving the optimization problem, the criterion of maximum substitution of diesel generation (or minimum

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diesel fuel consumption) and minimum total discounted costs is used while maintaining the balance of electric and thermal power at each moment in time [18].

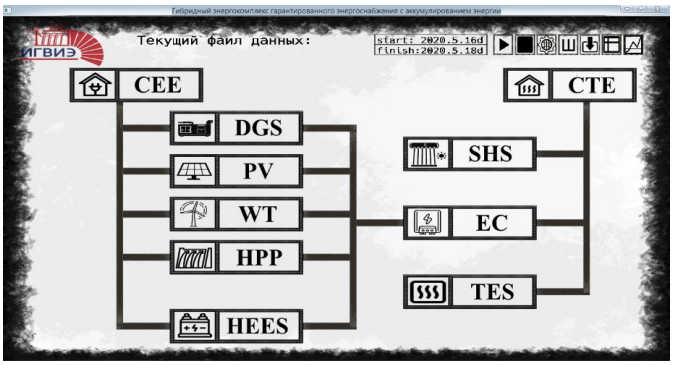


Fig. 3. RES Energy Complex Digital twin main control screen.

III. Development and Evaluation of Measures to Improve the Efficiency of HECs Using Verified Digital Twins

A. *Verification of digital twins using the example of the HEC in the village of Batagay*

Batagay is an urban-type settlement located in the northern part of the Republic of Sakha (Yakutia), Russian Federation in the floodplain on the right bank of the river Yana (geographical coordinates 67° 39′ N 134° 39′ E).

The power system of the village is isolated. The village has no industrial enterprises with significant energy consumption. Provision of consumers of the first, second and third reliability categories with electric energy is carried out by means of a hybrid energy complex (HEC) with a total installed capacity of 12 MW, consisting of 12 different diesel generator units (DGU), switched on as needed, and 3,360 multicrystalline photovoltaic units (PV) from the Chinese company Suntech Power, with a capacity of 300 W each. The main characteristics of the isolated power system of the Batagay village are given in Table 1.

TABLE I. The Main Characteristics of the Isolated Power System of the Batagay Village

Parameter name	Parameter value
Population, people	3 487
Minimum winter power of consumers, kW	3 275
Maximum winter power of consumers, kW	5 186
Minimum summer power of consumers, kW	907
Maximum summer power of consumers, kW	2 075
Installed capacity of DGU, kW	11 000
Electric energy consumption volume, MWh/year	23 526
Number of PV, pcs	3 360
Total installed capacity of PV, kW	1 000

Figure 4 shows a comparison of the actual operating data of the HEC in the village of Batagay for 2020 with the data calculated for the same year using the digital twins HOMER Pro, HOGA and RES Energy Complex.

The calculated data of all three digital twins exceed the actual operating data of the HEC in the village of Batagay, and the calculated data of HOGA and HOMER Pro have a large deviation from the actual data by 51% and 44%, respectively. Such large deviations can lead to a distortion of the design parameters of the HEC and to false expectations of investors.

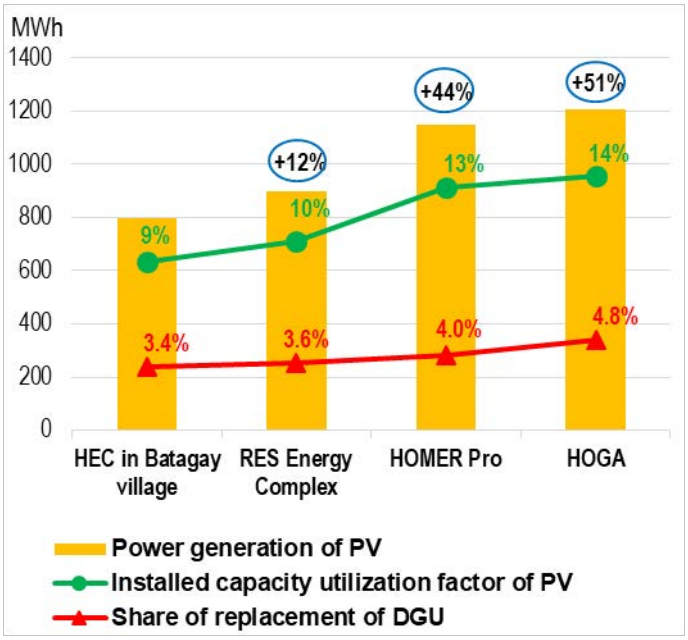


Fig. 4. Actual and estimated data on the operation of the HEC in the village of Batagay in 2020.

A comparison of the calculated values of the digital twin of RES Energy Complex with the corresponding actual values shows that the deviation of the calculated annual output of the PV as part of the HEC in the village of Batagay relative to its actual annual output is +12%. For HECs with renewable energy installations that have a diverse composition of equipment and a high dependence on many natural factors, this result can be considered acceptable.

B. *Verification of digital twins using the example of the HEC in the village of Novikovo*

Novikovo is a village in the Korsakovsky urban district of the Sakhalin region of the Russian Federation, located on the Tonino-Anivsky Peninsula on the shore of Aniva Bay. Electricity supply to consumers in the village is provided by a HEC with a total installed capacity of 5466 kW, which includes two VT-225 wind turbines (WT) (225 kW each) and a DGU with a capacity of 5016 kW. The main characteristics of the isolated power system of the Novikovo village are given in Table 2.

Comparison of actual data on the operation of the HEC in the village of Novikovo for 2020 with its calculated data obtained using the digital twins under consideration are presented in Figure 5.

The estimated value of the HOMER Pro digital twin WT output is 15 times higher than its actual output in 2020, with

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the actual and estimated values of the installed capacity factor of the WT being 3% and 45%, respectively.

TABLE II. THE MAIN CHARACTERISTICS OF THE ISOLATED POWER SYSTEM OF THE NOVIKOVO VILLAGE

Parameter name	Parameter value
Population, people	500
Minimum power of consumers, kW	107
Maximum power of consumers, kW	321
Installed capacity of Diesel Generator DGU , kW	5 016
Electric energy consumption volume, MWh/year	1 726
Number of Wind Turbines (WT), pcs	2
Total installed capacity of WT, kW	450

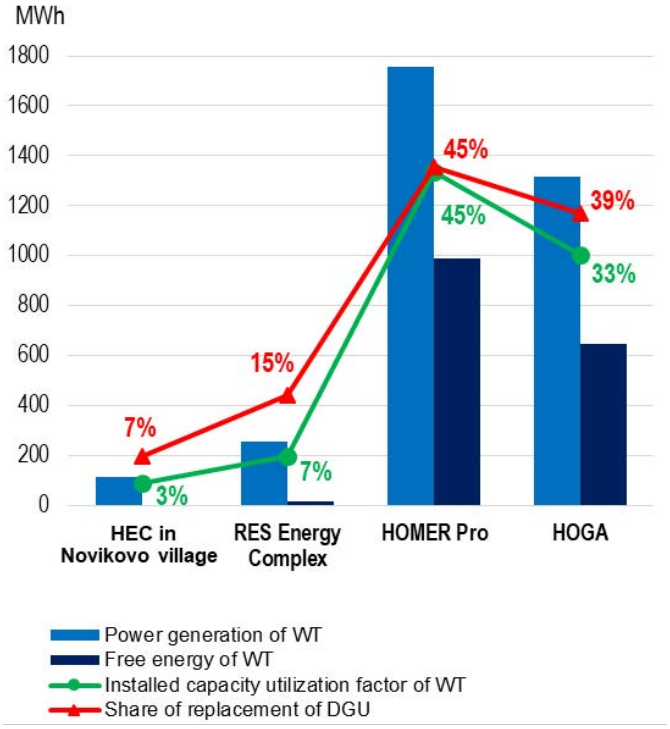


Fig. 5. Actual and estimated data on the operation of the HEC in the village of Novikovo in 2020.

The estimated value of the HOGA digital twin WT output exceeds its actual output in 2020 by 11.5 times with the estimated value of the installed capacity factor of the WT being 33%.

The calculated data of the digital twin of RES Energy Complex are the closest to the actual data of operation of the HEC in the village of Novikovo. The calculated value of the WT output is 2.2 times higher than its actual output in 2020 with the actual and calculated values of the installed capacity factor of the WT of 3% and 7%, respectively.

C. Evaluation of financial and economic indicators of the effectiveness of projects for the modernization of local generation facilities

In order to identify the need to develop and apply a set of measures to improve the efficiency of implementing projects to modernize local generation facilities in Russia, an assessment was made of the financial and economic indicators of 270 projects included in the Program for the Modernization of Local Generation Facilities in the Far East approved by the Government of the Russian Federation.

For the purposes of financial and economic modeling, the following key parameters were adopted:

- The project implementation period is 2 years.
- The main source of reimbursement of capital costs for the project is fuel savings as a result of the use of energy-efficient technical solutions.
- The rate of growth of fuel prices is 6.5% annually (based on statistical data).
- The average weighted discount rate is 14%.

Financial and economic modeling of the projects was carried out using the verified digital twin of RES Energy Complex Pro.

The summary calculation results are presented in Tables 3 and 4.

TABLE III. SUMMARY RESULTS OF THE ASSESSMENT OF FINANCIAL AND ECONOMIC INDICATORS OF PROJECTS OF THE 1ST STAGE OF THE PROGRAM

Indicators	Total objects	> 1 MW	0.5-1 MW	< 0.5 MW
	80	**18**	**13**	**49**
Fuel economy. tons/year	23 430	16 500	2 806	4 125
Capital expenditures. million rubles	21 424	10 942	2 865	7 618
Share of equity capital. %	20			
Share of borrowed funds. %	80			
Amount of borrowed funds. million rubles.	17 139			
Loan rate. %	25			
Loan term. years	10			
Calculation horizon. years	20			
Income tax. %	25			
Discount rate. %	14			
IRR. %	**5.6**	**11.7**	**3.7**	**-4.1**
Projects pay off. units.	**0**	**0**	**0**	**0**

TABLE IV. SUMMARY RESULTS OF THE ASSESSMENT OF FINANCIAL AND ECONOMIC INDICATORS OF PROJECTS OF THE 1ST STAGE OF THE PROGRAM

Indicators	Total objects	> 1 MW	0.5-1 MW	< 0.5 MW
	190	**45**	**31**	**114**
Fuel economy. tons/year	41 035	32 038	4 183	4 814
Capital expenditures. million rubles	21 424	10 942	2 865	7 618
Share of equity capital. %	20			
Share of borrowed funds. %	80			
Amount of borrowed funds. million rubles.	43 011			
Loan rate. %	25			
Loan term. years	10			
Calculation horizon. years	20			
Income tax. %	25			
Discount rate. %	14			
IRR. %	**0.1**	**6.4**	**-3.0**	**-11.0**
Projects pay off. units.	**0**	**0**	**0**	**0**

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The average weighted internal rate of return of energy service projects of the 1st stage of the Program implementation is only 5.6%, the 2nd stage of the Program – 0.1%. With the current rate of the Central Bank of the Russian Federation at 21% and the lack of state support mechanisms that provide access to cheaper borrowed capital, the feasibility of investing in projects to modernize local generation facilities remains low. High financial costs and risks associated with the implementation of such projects may lead to potential investors avoiding investments in such projects, which threatens not only their profitability, but also the development of the sector as a whole.

Thus, under current conditions, the economic efficiency of implementing the Program projects is possible only if the state provides additional support measures, including:

- subsidizing at least 30% of capital expenditures;
- provision of additional tax incentives;
- provision of preferential bank financing or interest rate subsidy mechanisms to ensure the attraction of borrowed funds at a level of 2–3%.

D. Development of a set of measures to support projects for the modernization of local generation facilities and an assessment of the effectiveness of their application

Based on the analysis of current models for attracting investment in the modernization and development of energy supply systems and mechanisms for state support for the implementation of infrastructure projects, a set of measures has been prepared for possible sources of co-financing for projects to modernize local generation facilities.

A numerical assessment of the effects of applying a set of measures was carried out using a verified digital twin of RES Energy Complex Pro using the example of projects of the 1st stage of the Program for the modernization of local generation facilities in the Far Eastern Federal District, taking into account the possible use of mechanisms after their testing for projects of the 2nd stage:

1. State support for projects for residents of the Arctic zone of the Russian Federation in accordance with the Decree of the Government of the Russian Federation of 21.12.2020 No. 2186.
2. State support for the implementation of priority investment projects with the participation of the state development corporation "VEB.RF" in accordance with the decrees of the Government of the Russian Federation dated 16.10.2014 No. 1055
3. State support for infrastructure projects in accordance with the Decree of the Government of the Russian Federation dated 31.12.2020 No. 2459.
4. Extension of the “Far Eastern supplement” mechanism until 2035 and use of part of the funds to finance the Program projects.
5. Changes in regulatory legal acts in terms of expanding cost savings, which can be preserved in the tariff of the regulated organization.
6. Support for technology companies in order to refine the products of technology companies to meet the requirements of large corporations (approved by order of the autonomous non-profit organization "Engineering and Innovation Support Center" dated 30.06.2023 No. 07/6).
7. Exemption of investors from paying income tax during the project payback period.

The results of calculating the efficiency of using each of the listed mechanisms separately are summarized in Table 5.

TABLE V. SUMMARY RESULTS OF THE ASSESSMENT OF FINANCIAL AND ECONOMIC INDICATORS OF PROJECTS OF THE 1ST STAGE OF THE PROGRAM

Indicators	**Current conditions**	**Mechanism 1**	**Mechanism 2**	**Mechanism 3**	**Mechanism 4**	**Mechanism 5**	**Mechanism 6**	**Mechanism 7**
IRR. %	5.6	9.0	8.4	11.7	9.9	15	8	6.2
Budget funds. million rubles	–	7636	6504	12255	9427	–	–	197
Annual budget funds. million rubles	–	763.6	650.4	1226	943	–	–	98
Duration of support measures. years	–	10	10	10	10	10	–	2
Loan term. years	10							
Market rate on the loan. %	25	25	25	–	–	25	25	25
Preferential loan rate. %	–	16	2	12	15	–	–	–
Average weighted interest rate on a loan. %	25	16.9	18.1	12	15	25	25	25
Share of equity capital. %	20	20	20	20	20	50	20	20
Share of borrowed funds. %	80	80	80	80	80	50	80	80
Calculation horizon. years	20							
Discount rate, %	14							

IV. CONCLUSION

In the current macroeconomic conditions, the economic efficiency of implementing projects to create hybrid energy complexes with renewable energy power plants is possible only if the state provides additional support measures, such as subsidizing at least 30% of capital costs or providing additional tax incentives and preferential bank financing or interest rate subsidy mechanisms that ensure the attraction of borrowed funds at a level of 2–3%.

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To achieve stable efficiency, it seems advisable to combine projects with high and low efficiency into separate groups for implementation within the framework of one energy service project, which will optimize financial results and minimize risks.

The availability of a verified universal and multifunctional digital twin allows for a quick detailed economic analysis of different options for energy complexes with renewable energy power plants at the project initiation stage to eliminate possible risks, as well as to assess the impact of possible support measures on project parameters.

A numerical assessment of the implementation of the developed set of support measures, carried out using the verified RES Energy Complex digital twin based on data from real projects, confirms their effectiveness and efficiency.

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